

A SIDON BASIS OF ORDER THREE FROM POLYNOMIALS OVER $\mathbb{F}_2$

ABSTRACT. We give an elementary proof of Pilatte’s theorem that an infinite Sidon set can be an asymptotic basis of order three. The construction uses irreducible polynomials over the binary field. We replace distribution of triple products in every class modulo a large polynomial by distribution of primes modulo a much smaller divisor. Auxiliary tags allow random shifts of the target residue class without changing the Sidon property. The tags have polynomial size, so their cost is negligible, and one joint random choice gives the basis property.

1. INTRODUCTION

A set S of positive integers is *Sidon* if $a + b = c + d$, with $a, b, c, d \in S$, implies equality of the unordered pairs $\{a, b\}$ and $\{c, d\}$, counted with multiplicity. It is an *asymptotic basis of order three* if every sufficiently large integer is a sum of three elements of S . Pilatte [Pil24] proved that these properties are compatible, answering a question of Erdős, Sárközy, and Sós.

Theorem 1.1. *There is an infinite Sidon set of positive integers that is an asymptotic basis of order three.*

We give a proof using polynomial arithmetic, finite character sums, and the probabilistic method. The only distribution estimate needed comes from an elementary zero-free region for polynomial Dirichlet L -functions. In particular, the proof uses neither the Riemann hypothesis for curves nor estimates for factorization functions in progressions of large modulus.

1.1. Comparison with Pilatte’s proof. Pilatte encodes irreducible polynomials by their discrete logarithms modulo a sequence of auxiliary irreducibles. Equality of two sums of encoded integers forces polynomial congruences, and degree comparisons give the Sidon property. To represent a prescribed integer as a sum of three terms, his construction requires many triple products in one prescribed class modulo the full product of the auxiliary moduli. His Lemma 5.2 obtains this from Sawin’s estimate for convolutions of the von Mangoldt function [Saw24, Lemma 9.14].

We retain the polynomial encoding and its degree comparisons. The change is to let the target class vary. In each block we attach a tag t from a finite field of characteristic seven, record both t and t^2 , and add a shared random function of t to the discrete logarithm. The sum and sum of squares determine an unordered pair of tags. Thus the shifts cancel whenever two pair sums agree. For triples, the same two moments leave many choices, which give independent attempts to reach a class with many prime products.

Consequently, we need only a small proportion of such classes above each class of a short divisor. This follows by counting triple products modulo the divisor and using the trivial upper bound for their number in a class of the full modulus. There is no cancellation estimate at the full modulus. The extra freedom in the construction is what permits the weaker arithmetic input.

1.2. Outline of the proof. Put $\gamma = 7/20$. At stage k , the full modulus, denoted by g_k , has degree k^2 . The labels are monic irreducibles of an even degree $d_k = \gamma k^2 + O(1)$. Thus the ratio of the modulus degree to the triple-product degree tends to $20/21$. The inequality $3\gamma > 1$ makes the average number of triple products per residue class exponential in k^2 . The Sidon argument requires the inequality $(1 - \gamma)^2 > \gamma$, which also holds. These conditions leave the interval

$$\frac{1}{3} < \gamma < \frac{3 - \sqrt{5}}{2}.$$

For the arithmetic input, take the divisor g_h with $h = \lfloor k^{1/4} \rfloor + 1$. Its degree is $h^2 = O(k^{1/2})$. An elementary character estimate gives primes of degree d_k in every invertible class modulo g_h . It follows that, above each such class, a proportion at least $1/(1024d_k^3)$ of the invertible classes modulo g_k contain at least

$$\frac{2^{3d_k - k^2}}{1024d_k^3}$$

unordered triples of distinct primes. We call these classes good.

The tag field in block i has size comparable to $(i+1)^{40}$. For prescribed first and second moments, the blocks with $i \geq h$ then supply k^{10} disjoint tag trials. We use the same tags in the blocks with $i < h$ and condition on their shifts. Each trial selects an independent uniform extension of the resulting class modulo g_h . The probability of missing all good classes is at most $\exp(-k^4/1024)$.

It matters that the good-class estimate holds above *every* short-prefix class. The low blocks are too small to supply all the independent trials; their shifts must be fixed before the high blocks are sampled. A density statement for the full group alone would not control this conditional distribution.

Finally, sample the parameters attached to individual primes, independently of the shifts. Distinct triples in a fixed residue class share no prime, by unique factorization and $2d_k < k^2$. Their attempts to represent the target are therefore independent. A prime has only $\exp(O(k \log k))$ possible parameter choices, whereas a good class contains $\exp(\Omega(k^2))$ triples. The resulting failure probabilities remain summable after counting every integer in each stage. We sample all parameters in one product space and apply Borel–Cantelli once.

All logarithms without a subscript are natural. Implied constants are absolute. We use $\mathbb{N} = \{0, 1, 2, \dots\}$; the constructed set will lie in $\mathbb{N} \setminus \{0\}$.

2. THE ELEMENTARY DISTRIBUTION ESTIMATE

Write $\mathcal{M}_n$ and $\mathcal{P}_n$ for the monic polynomials and monic irreducible polynomials of degree n in $\mathbb{F}_2[t]$. For a monic polynomial g , put

$$\phi(g) = |(\mathbb{F}_2[t]/g)^\times|, \quad \pi_n(a; g) = |\{P \in \mathcal{P}_n : P \equiv a \pmod{g}\}| \quad (a \in (\mathbb{F}_2[t]/g)^\times).$$

Lemma 2.1. *Let g be monic of degree $H \geq 1$, and suppose $\phi(g)$ is odd. Uniformly in $a \in (\mathbb{F}_2[t]/g)^\times$ and integers $n > H$,*

$$(2.1) \quad |n\phi(g)\pi_n(a; g) - 2^n| \ll \phi(g) \left(H2^n e^{-n/(100H)} + n^2 2^{n/2} \right).$$

In particular, if $n \rightarrow \infty$ and $H^2 = o(n)$, then $\pi_n(a; g) \sim 2^n/(n\phi(g))$, uniformly in g and a .

Proof. Extend each character χ of $(\mathbb{F}_2[t]/g)^\times$ by zero on polynomials not coprime to g . For non-principal χ , define

$$L(z, \chi) = \sum_{f \text{ monic}} \chi(f) z^{\deg f} = \prod_j (1 - \alpha_j z).$$

Character orthogonality in each degree at least H shows that this is a polynomial of degree at most $H - 1$. Its Euler product is nonvanishing for $|z| < 1/2$, so $|\alpha_j| \leq 2$. We claim that

$$(2.2) \quad |\alpha_j| < 2 \left(1 - \frac{1}{100H} \right).$$

Since the character group has odd order, χ^2 is nonprincipal. Put

$$S_\chi(z) = \frac{zL'(z, \chi)}{L(z, \chi)}.$$

Applying $3 + 4 \cos \theta + \cos 2\theta \geq 0$ termwise to the Euler logarithmic derivatives gives

$$(2.3) \quad 0 \leq \frac{6\rho}{1-2\rho} + 4 \operatorname{Re} S_\chi(\rho e^{i\theta}) + \operatorname{Re} S_{\chi^2}(\rho e^{2i\theta}) \quad (0 < \rho < 1/2).$$

Primes dividing g cause no difficulty: their contribution to the first term is nonnegative. Also $\operatorname{Re}(-w/(1-w)) \leq 1/2$ for $|w| < 1$.

If an inverse root α violated (2.2), take $x = 1 - 1/(10H)$, $\rho = x/2$, and choose θ so that $y = \alpha \rho e^{i\theta}$ is real and positive. Then $1 > y \geq 1 - 11/(100H)$. The contribution of this root to S_χ is $-y/(1-y)$; all other root contributions are at most $1/2$. Since $H \geq 2$ if there is a root, the right side of (2.3) is at most

$$\frac{3x}{1-x} - \frac{4y}{1-y} + \frac{5H-9}{2} < 0,$$

a contradiction. This proves (2.2).

Comparison of coefficients in the logarithmic derivative gives

$$\sum_{\ell \deg P = n} \deg(P) \chi(P)^\ell = - \sum_j \alpha_j^n,$$

where the sum on the left is over monic irreducibles P and integers $\ell \geq 1$. The terms with $\ell \geq 2$ contribute $O(n^2 2^{n/2})$. For the principal character, use $\sum_{f \text{ monic}} z^{\deg f} = (1-2z)^{-1}$ and remove the same prime-power terms. Since $n > H$, no prime of degree n divides g . Character orthogonality now proves (2.1). Its relative error tends to zero when $H^2 = o(n)$, since $\phi(g) \leq 2^H$. $\square$

3. MANY GOOD CLASSES ABOVE EACH SHORT PREFIX

Choose $p_i \in \mathcal{P}_{2i+1}$ for every $i \geq 0$, and define

$$g_k = \prod_{i < k} p_i, \quad G_k = (\mathbb{F}_2[t]/g_k)^\times, \quad \phi_k = |G_k|, \quad N_i = 2^{2i+1} - 1.$$

The polynomials p_i are distinct. Thus g_k is squarefree, $\deg g_k = k^2$, and $\phi_k = \prod_{i < k} N_i \leq 2^{k^2}$ is odd. Fix group isomorphisms

$$\lambda_i : (\mathbb{F}_2[t]/p_i)^\times \longrightarrow \mathbb{Z}/N_i\mathbb{Z}.$$

By the Chinese remainder theorem, their product identifies G_k with $\prod_{i < k} \mathbb{Z}/N_i\mathbb{Z}$.

Set

$$(3.1) \quad \gamma = \frac{7}{20}, \quad d_k = 2 \left\lceil \frac{7k^2}{40} \right\rceil, \quad h_k = \lfloor k^{1/4} \rfloor + 1.$$

For large k , the degrees d_k are strictly increasing and

$$(3.2) \quad d_k = \gamma k^2 + O(1), \quad 2d_k < k^2 < 3d_k.$$

Every prime of degree d_k is coprime to every p_i , since their degrees have opposite parity. We henceforth take stages large enough that these inequalities hold.

Fix a stage k , and abbreviate $d = d_k$ and $h = h_k$. For $v \in G_k$, let $\mathcal{T}_k(v)$ be the set of unordered triples of distinct elements of $\mathcal{P}_d$ whose product is congruent to v modulo g_k . Define

$$U_k = 2^{3d-k^2}, \quad \varepsilon_k = \frac{1}{1024d^3}, \quad T_k = \varepsilon_k U_k.$$

We call v *good* if $|\mathcal{T}_k(v)| \geq T_k$.

Lemma 3.1. *For all sufficiently large k , at least an ε_k fraction of the classes in each fiber of $G_k \rightarrow G_h$ are good. Moreover, distinct triples in any $\mathcal{T}_k(v)$ are disjoint.*

Proof. We have $(\deg g_h)^2 = h^4 = O(k) = o(d)$. By Lemma 2.1, each class in G_h contains at least

$$L = \frac{2^d}{2d\phi_h}$$

primes of degree d , for all sufficiently large k . In particular, $L \rightarrow \infty$.

Fix $u \in G_h$, and put $M = |\mathcal{P}_d|$. For each ordered pair of primes, there are at least L choices of a third prime making their product congruent to u . At most $3M^2$ ordered triples have a repeated entry. Dividing by six and using $M \geq \phi_h L$ therefore gives

$$(3.3) \quad \sum_{v \mapsto u} |\mathcal{T}_k(v)| \geq \frac{2^{3d}}{512d^3\phi_h}.$$

Unique factorization identifies a triple with its product. A class modulo g_k contains exactly $2^{3d-k^2} = U_k$ monic polynomials of degree $3d$, so $|\mathcal{T}_k(v)| \leq U_k$. Let $N = \phi_k/\phi_h$ be the number of extensions of u , and let J of them be good. The left side of (3.3) is at most $NT_k + JU_k$, while its right side is at least $2NT_k$. Hence $J \geq N\varepsilon_k$.

If two triples in $\mathcal{T}_k(v)$ share a prime, cancel it modulo g_k . The products of the remaining pairs are congruent modulo g_k and have degree $2d < k^2$, so they are equal. Unique factorization identifies the triples. $\square$

4. TAGS AND THE INTEGER ENCODING

For a tuple of field elements, its first and second moments mean the sum of its entries and the sum of their squares. We use the following property of characteristic seven.

Lemma 4.1. *Let E be a finite field of characteristic seven. The first two moments determine an unordered pair in E . For any $u, v \in E$ and integer $r \geq 1$ with $|E| \geq 7r$, there are r nonconstant triples with moments u, v whose sets of entries are pairwise disjoint.*

Proof. The pair assertion follows from $2ab = (a + b)^2 - (a^2 + b^2)$. For each $x \in E^\times$, put $y = (v + 2u^2)/x$. The triple

$$(3x + 2y + 5u, -x + y - 2u, -2x - 3y - 2u)$$

has moments u, v , is nonconstant, and determines x . Thus there are at least $|E| - 1$ such ordered triples. If two triples with the same moments share an entry, the pair assertion identifies their remaining entries. Distinct supports are therefore disjoint, and each support accounts for at most six ordered triples. Since $(|E| - 1)/6 \geq r$, the result follows. $\square$

For block $i \geq 0$, define

$$(4.1) \quad r_i = 1 + 40 \lceil \log_7(i + 1) \rceil, \quad E_i = \mathbb{F}_{7^{r_i}}.$$

Fix an $\mathbb{F}_7$ -basis of each E_i . We identify its coordinates with integers in $\{0, \dots, 6\}$. Notice that

$$|E_i| \geq 7(i + 1)^{40}, \quad r_i = O(\log(i + 2)).$$

The auxiliary digit set is

$$A = \{10, 13, 14, 15, 16, 17, 22, 41, 42, 43, 46, 47, 48, 49, 50\}.$$

It satisfies

$$(4.2) \quad A \cap (A + A + \{0, 1\}) = \emptyset, \quad [37, 141] \cap \mathbb{Z} \subseteq A + A + A.$$

These are finite checks. For an ordinary digit x , with $0 \leq x < b$ and integer $b \geq 1$, we encode x together with an auxiliary digit $a \in A$ as $x + ba$ in radix $103b$. Every encoded digit is less than half its radix, so pair sums have no carries between encoded positions. The first property in (4.2) also prevents a single encoded digit from equaling the sum of two.

Choose functions, called *masks*,

$$\tau_i : E_i \longrightarrow \mathbb{Z}/N_i\mathbb{Z} \quad (i \geq 0).$$

A prime $P \in \mathcal{P}_{d_k}$ is a label at stage k . Write $\lambda_i(P)$ for the logarithm of its residue modulo p_i . For each $i < k$, choose a tag $t_i(P) \in E_i$. The ordinary data in its i -th block are

$$(4.3) \quad \lambda_i(P) + \tau_i(t_i(P)), \quad t_i(P), \quad t_i(P)^2.$$

Use the representative in $\{0, \dots, N_i - 1\}$ for the first entry and the chosen base-seven coordinates for the last two. Encode each ordinary digit with its own choice from A .

Read these positions in the displayed order, from least to most significant. Let $C_i(P)$ be the resulting block value. Its radix and the successive block place values are

$$(4.4) \quad R_i = 103N_i 721^{2r_i}, \quad B_0 = 1, \quad B_k = \prod_{i < k} R_i.$$

Choose also a top digit $s(P) \in \{1, \dots, 2^{3k}\}$, and put

$$(4.5) \quad n(P) = \sum_{i < k} B_i C_i(P) + B_k s(P).$$

For a sufficiently large fixed integer k_0 , let S be the set of these integers over all stages $k \geq k_0$.

The half-radix bound gives

$$(4.6) \quad 2C_i(P) < R_i, \quad 2 \sum_{i < j} B_i C_i(P) < B_j \quad (j \leq k).$$

We will also use the elementary growth bounds

$$(4.7) \quad R_i \geq 2^{2i+1}, \quad \log_2 R_i = 2i + O(\log(i+2)),$$

$$(4.8) \quad B_k \geq 2^{k^2}, \quad \log_2 B_k = k^2 + O(k \log k),$$

$$(4.9) \quad B_k \leq n(P) < (2^{3k} + 1)B_k, \quad 2n(P) < B_{k+2}.$$

For the last inequality, two further block radices have product at least $2^{4k+4} > 4 \cdot 2^{3k}$.

The number of choices of tags, auxiliary digits, and top digit for one prime at stage k is

$$(4.10) \quad D_k = 2^{3k} \prod_{i < k} 7^{r_i} 15^{1+2r_i} = 2^{O(k \log k)}.$$

This estimate is the reason for using polynomial-sized tag fields. Their total contribution to the exponent is smaller than the quadratic excess $3d_k - k^2$, even over $\mathbb{F}_2$.

5. THE SIDON PROPERTY

Proposition 5.1. *For sufficiently large k_0 , every choice of masks and every choice of parameters for the primes produces a Sidon set S .*

Proof. Equality of individual blocks recovers their tags, then their masked logarithms, and hence the residues of their labels modulo p_i . Equality of pair sums of blocks recovers the two moments of the tag pairs. By Lemma 4.1, the unordered pairs agree, so the mask sums cancel. The products of the two labels are therefore congruent modulo p_i . These conclusions hold for arbitrary masks.

Suppose

$$n(P_1) + n(P_2) = n(P_3) + n(P_4),$$

with stages $a \geq b$ on the left and $c \geq e$ on the right. The bounds (4.9) imply $|a - c| \leq 1$. Also $|b - e| \leq 3$. Indeed, if $b \geq e + 4$, block $e + 2$ compares two complete blocks on the left with one on the right. The shorter summand on the right, together with the lower part of the longer summand, is less than B_{e+2} by (4.6) and (4.9). It creates no carry into that block. This contradicts the single-versus-pair exclusion in (4.2). The other direction is symmetric.

Assume the equality is nontrivial. Unique factorization gives $P_1 P_2 \neq P_3 P_4$, and cancellation gives $P_1 \neq P_3$. Comparison of the common lower blocks yields

$$g_{\min(b,e)} \mid P_1 P_2 - P_3 P_4.$$

Comparison above the two shorter summands yields

$$\prod_{\max(b,e)+2 \leq i < \min(a,c)} p_i \mid P_1 - P_3.$$

An empty product in the second line is interpreted as 1. Using the degree bounds and the closeness of the stages, these two divisibilities imply, respectively,

$$b^2 \leq \gamma(a^2 + b^2) + O(a), \quad a^2 - b^2 \leq \gamma a^2 + O(a).$$

Thus

$$(1 - \gamma)a^2 + O(a) \leq b^2 \leq \frac{\gamma}{1 - \gamma}a^2 + O(a).$$

Since $(1 - \gamma)^2 > \gamma$, this is impossible for large a . Choosing k_0 large enough proves the proposition. $\square$

6. A JOINT RANDOM CHOICE

We now choose each mask τ_i uniformly from all functions with its specified domain and codomain. Independently, for each prime label choose its tags, auxiliary digits, and top digit uniformly from the D_k possibilities at its stage. All these choices are mutually independent. They define one countable product probability space. We write $\mathbb{P}_\tau$ for probability over the masks alone.

6.1. Moving the target to a good class. Fix a sufficiently large stage k . A *moment target* consists of elements

$$x_i \in \mathbb{Z}/N_i\mathbb{Z}, \quad u_i, v_i \in E_i \quad (i < k).$$

It is *hit* by the masks if there are triples $(t_{i,1}, t_{i,2}, t_{i,3})$ with moments u_i, v_i such that the vector

$$(6.1) \quad \left(x_i - \sum_{j=1}^3 \tau_i(t_{i,j}) \right)_{i < k}$$

is the logarithm vector of a good class in G_k .

Lemma 6.1. *For every moment target at every sufficiently large stage k ,*

$$\mathbb{P}_\tau(\text{the target is not hit}) \leq \exp(-k^4/1024).$$

Proof. Put $h = h_k$ and $r = k^{10}$. For each block $i < h$, choose one nonconstant triple with the required moments and use it in all r trials. For $h \leq i < k$, choose r triples with those moments and disjoint supports. Lemma 4.1 allows this, since $|E_i| \geq 7(i+1)^{40} \geq 7k^{10}$.

Condition on the masks in the blocks $i < h$. The low coordinates of (6.1) now specify a class $u \in G_h$. A nonconstant triple has an entry occurring exactly once, so the sum of its three mask values is uniform in $\mathbb{Z}/N_i\mathbb{Z}$. Disjoint supports give independent sums. Thus the r trials are independent uniform extensions of u to G_k . Lemma 3.1 bounds their simultaneous failure probability by

$$(1 - \varepsilon_k)^r \leq \exp\left(-\frac{k^{10}}{1024d_k^3}\right) \leq \exp(-k^4/1024),$$

since $d_k \leq k^2$. The bound is uniform in the low masks, so it also holds without conditioning. $\square$

6.2. Representing an integer once its target is hit. Define the stage window

$$I_k = [6B_k, 6B_{k+1}) \cap \mathbb{Z}.$$

Fix $m \in I_k$. Successive division in the encoded radices gives a representative D of m modulo B_k , with $0 \leq D < 2B_k$, whose digit at each ordinary base b has the form

$$x + by, \quad 0 \leq x < b, \quad 39 \leq y \leq 141.$$

Indeed, the permitted digits form the interval $[39b, 142b - 1] \cap \mathbb{Z}$, a complete residue system modulo $103b$; the bound on D follows from their sizes. Setting $z = (m - D)/B_k$, we obtain

$$(6.2) \quad m = D + B_k z, \quad 4 \leq z < 6R_k.$$

By (4.7), $2R_k \leq 2^{3k}$ for all sufficiently large k , so

$$(6.3) \quad 3 \leq z \leq 3 \cdot 2^{3k}.$$

Fix one such expansion for each m , before choosing the masks.

The ordinary digits of D give a moment target: its first coordinate in block i gives x_i , and its two groups of base-seven coordinates give u_i, v_i . Suppose the masks hit this target. Choose witnessing tag triples and a good class v before revealing any of the parameters attached to primes.

Every triple $\{P_1, P_2, P_3\} \in \mathcal{T}_k(v)$ then has a parameter choice for which

$$(6.4) \quad n(P_1) + n(P_2) + n(P_3) = m.$$

To see this, order the three primes arbitrarily and assign the witnessing tags to them. The product congruence and the two moment identities give the ordinary target digits modulo their bases. At any position, if the three ordinary digits are x_1, x_2, x_3 , we have

$$x_1 + x_2 + x_3 = x + b\kappa, \quad \kappa \in \{0, 1, 2\}.$$

By (4.2), choose $a_1, a_2, a_3 \in A$ with $a_1 + a_2 + a_3 = y - \kappa$. The three encoded digits now sum to $x + by$ exactly. Finally choose three top digits with sum z , using (6.3). This proves (6.4).

Conditional on the masks, one prescribed parameter assignment to three distinct primes has probability D_k^{-3} . The triples in $\mathcal{T}_k(v)$ are disjoint by Lemma 3.1, so these events are independent. Since v is good, the conditional probability that all these representations fail is at most

$$(6.5) \quad (1 - D_k^{-3})^{|\mathcal{T}_k(v)|} \leq \exp(-T_k/D_k^3).$$

From (3.2) and (4.10),

$$\log_2(T_k/D_k^3) = 3d_k - k^2 - O(k \log k) = \frac{k^2}{20} - O(k \log k).$$

In particular, $T_k/D_k^3 \geq 2^k$ for large k . Combining (6.5) with Lemma 6.1 gives the unconditional bound

$$(6.6) \quad \mathbb{P}(m \text{ has no representation using stage } k) \leq e^{-k^4/1024} + e^{-2^k} \quad (m \in I_k).$$

6.3. Eventual coverage.

Proof of Theorem 1.1. By (4.8), there is an absolute constant $C > 0$ such that $|I_k| \leq \exp(Ck^2)$ for large k . The union bound and (6.6) give

$$\mathbb{P}(\text{some } m \in I_k \text{ is not represented at stage } k) \leq e^{Ck^2} \left(e^{-k^4/1024} + e^{-2^k} \right).$$

The right side is summable over k . Borel–Cantelli therefore shows that, almost surely, every sufficiently large window is covered. No independence between windows is needed. Since the windows are consecutive and $B_k \rightarrow \infty$, S is almost surely an asymptotic basis of order three.

Proposition 5.1 gives the Sidon property for every realization. The set is infinite because a finite set has a bounded triple sumset. Taking any realization with eventual coverage proves the theorem. $\square$

7. A VARIANT OVER A LARGER FIELD

The proof above follows the construction formalized as **Erdos157b**. The construction **Erdos157** instead uses $\mathbb{F}_{2^{1024}}$ for the polynomial coefficients, tag fields $\mathbb{F}_{7^{i+2}}$, prefix length $12\lceil \log_7 k \rceil + 1$, and k^{12} mask trials. It first chooses masks that work for all moment targets at every sufficiently large stage, then chooses the parameters attached to primes; the two choices use separate Borel–Cantelli arguments. The larger coefficient field pays for the quadratic tag cost. The short-prefix prime estimate, the good-class argument, and the degree comparisons proving the Sidon property are unchanged in principle.

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